

1) **Set the target.** Define a theoretical estimand.

Requires substantive **argument**.

Average difference in the **potential outcome** each woman i would realize

$$\tau = \frac{1}{n} \sum_{i=1}^n \left(\overbrace{Y_i(\text{Mother}, \text{Employed})}^{\text{if she were an employed mother}} - \overbrace{Y_i(\text{Non-mother}, \text{Employed})}^{\text{if she were an employed non-mother}} \right)$$

2) **Link to observables.** Define an empirical estimand under a certain identification design.

Requires conceptual **assumptions**.

Average difference in the **realized outcomes** of women with the covariates $\bar{x}_i$ of women i who

$$\theta = \frac{1}{n} \sum_{i=1}^n \left(\overbrace{\mathbb{E} \left(Y \mid \begin{array}{l} \text{Motherhood} = \text{Mother} \\ \text{Employment} = \text{Employed} \\ \text{Covariates } \bar{X} = \text{Observed } \bar{x}_i \end{array} \right)}^{\text{actually are mothers}} - \overbrace{\mathbb{E} \left(Y \mid \begin{array}{l} \text{Motherhood} = \text{Non-mother} \\ \text{Employment} = \text{Employed} \\ \text{Covariates } \bar{X} = \text{Observed } \bar{x}_i \end{array} \right)}^{\text{actually are not mothers}} \right)$$

3) **Learn from data.** Select an estimation strategy.

Requires statistical **evidence**.

Average difference in the **regression prediction** at the covariates $\bar{x}_i$ of women i if

$$\underbrace{\hat{\theta}}_{\text{estimate of the estimand}} = \frac{1}{n} \sum_{i=1}^n \left(\underbrace{\widehat{\mathbb{E}} \left(Y \mid \begin{array}{l} \text{Motherhood} = \text{Mother} \\ \text{Employment} = \text{Employed} \\ \text{Covariates } \bar{X} = \text{Observed } \bar{x}_i \end{array} \right)}_{\text{estimated } \widehat{Y}_i(\text{Mother})} - \underbrace{\widehat{\mathbb{E}} \left(Y \mid \begin{array}{l} \text{Motherhood} = \text{Non-mother} \\ \text{Employment} = \text{Employed} \\ \text{Covariates } \bar{X} = \text{Observed } \bar{x}_i \end{array} \right)}_{\text{estimated } \widehat{Y}_i(\text{Non-mother})} \right)$$